

ECSE-4040

Digital Electronics

Lecture 28

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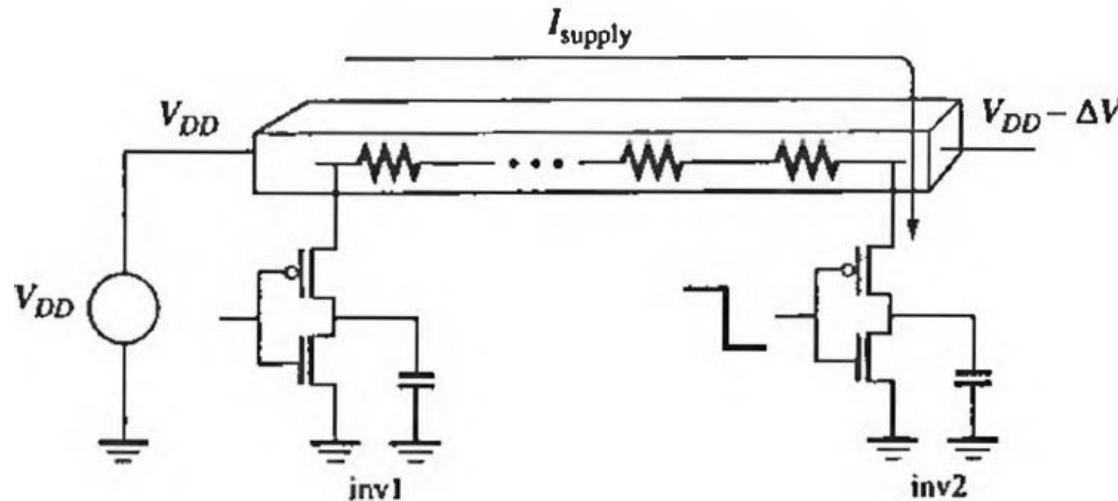
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<https://lms.rpi.edu>

Lecture Outline

- Power Grid and Clock Design
 - Power Distribution Design
 - Clocking and Timing Issues
 - Phase-Locked Loops/Delay-Locked Loops

Power Distribution Design



- Voltage noise in the supply is budgeted at $\pm 10\%$ X (V_{DD} — Gnd) as a rule of thumb
- Target Impedance
$$Z = \frac{V_{DD} \times (\text{fractional noise budget})}{I_{DD}}$$
- With $V_{DD} = 1.2V$ and $I_{DD} = 100A$,
$$Z = 1.2V \times (10\%) / 100A = 1.2m\Omega$$

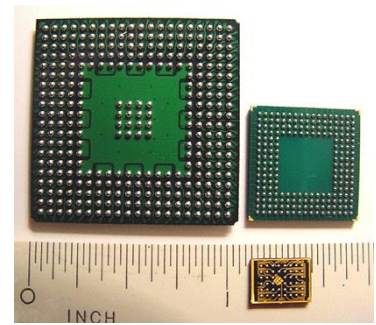
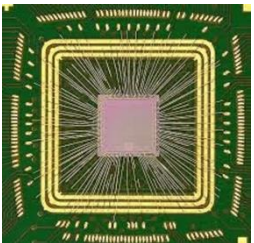
Interconnect Issues

- IR Drop and Ldi/dt
 - Total voltage drop at any point in power grid

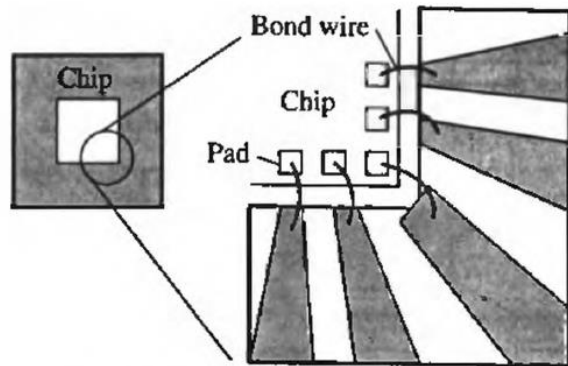
$$V = IR + L \frac{di}{dt}$$

- Package Pin Inductance

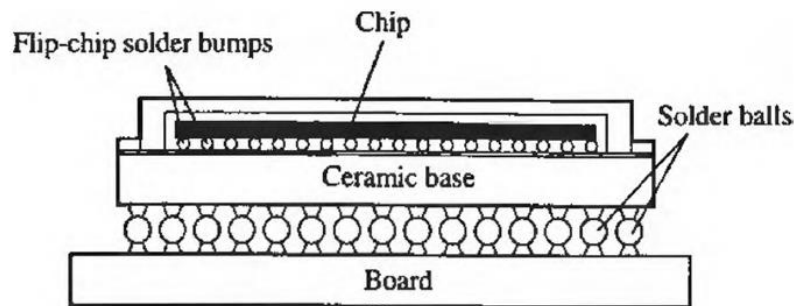
- Typically 1-2nH
- L has not increased but di/dt has
- Bonding wires vs. Ball Grid Array (BGA)



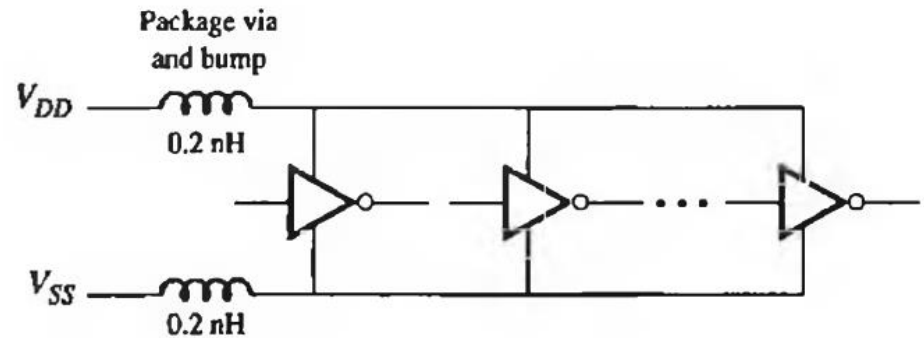
Chip Packaging and Parasitic Inductance



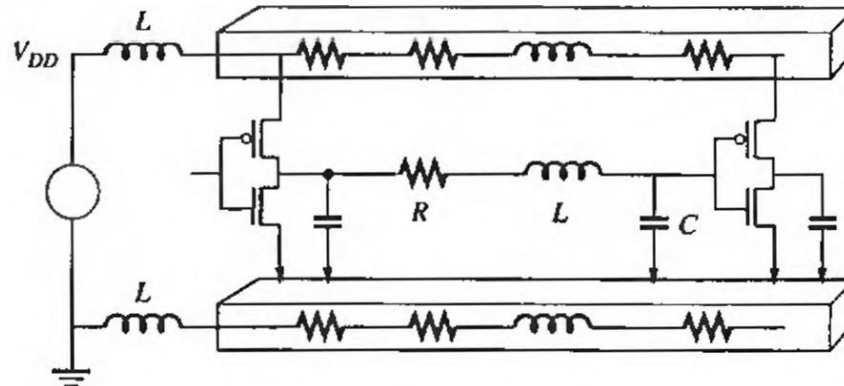
(a) Dual-inline packaging



(b) Ball grid array packaging



Chip Packaging and Parasitic Inductance

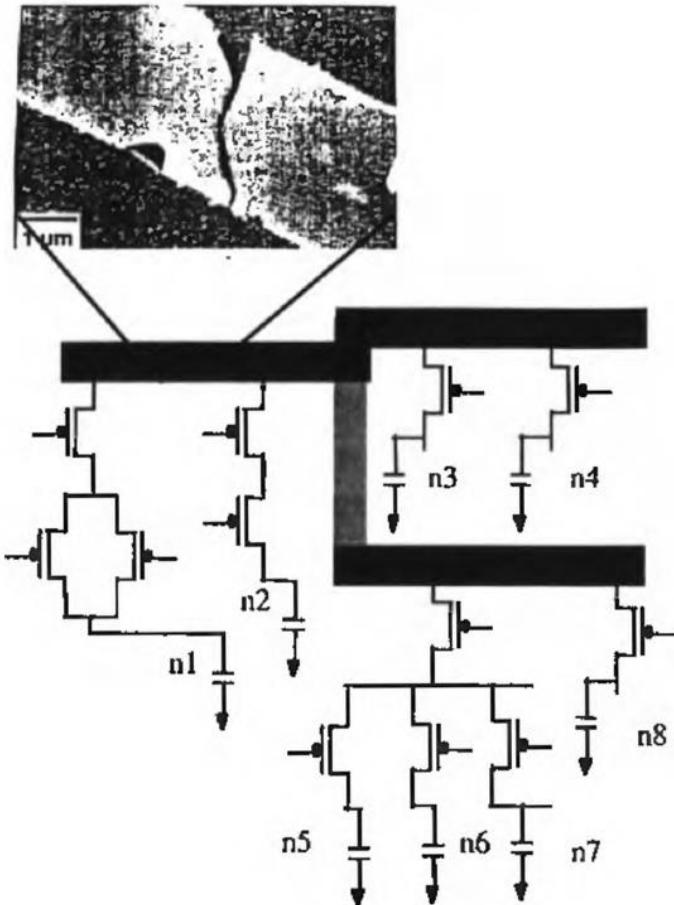


- If $L=0.2\text{nH}$ per bump and via,
Total Voltage Drop

$$V_L = 2 \times L \frac{di}{dt} = 2 \times 0.2 \text{ nH} \times \frac{25 \text{ mA}}{100 \text{ ps}} = 100 \text{ mV}$$

Since $V_{DD} = 1.2\text{V}$, 10% voltage drop!

Electromigration



- Thermally activated, current density inducing aluminum diffusion process, leading to voids in metal interconnect lines
- Process solution: Adding a small percentage ($\sim 1\%$) of copper in aluminum

$$\text{MTTF} = \frac{A}{J^2} \exp(\Delta H/kT)$$

HJS Ex. 11.1

Use of MTTF to Establish Maximum Allowable Current Density

Problem:

A design has a required lifetime of 10 years. Measurements for Al electromigration using an accelerated testing method indicates that $A = 2 \times 10^7$ hr-cm²/ampere and $\Delta H = 0.85$ eV. What is the maximum allowable current density in Al interconnect at 125 °C?

Solution:

The MTTF is the time at which 50% of the parts will fail, that is, t_{50} . For a desired lifetime of 10 years, the MTTF should be 5-10 times larger than this value. We will use a factor of 10 to be on the safe side:

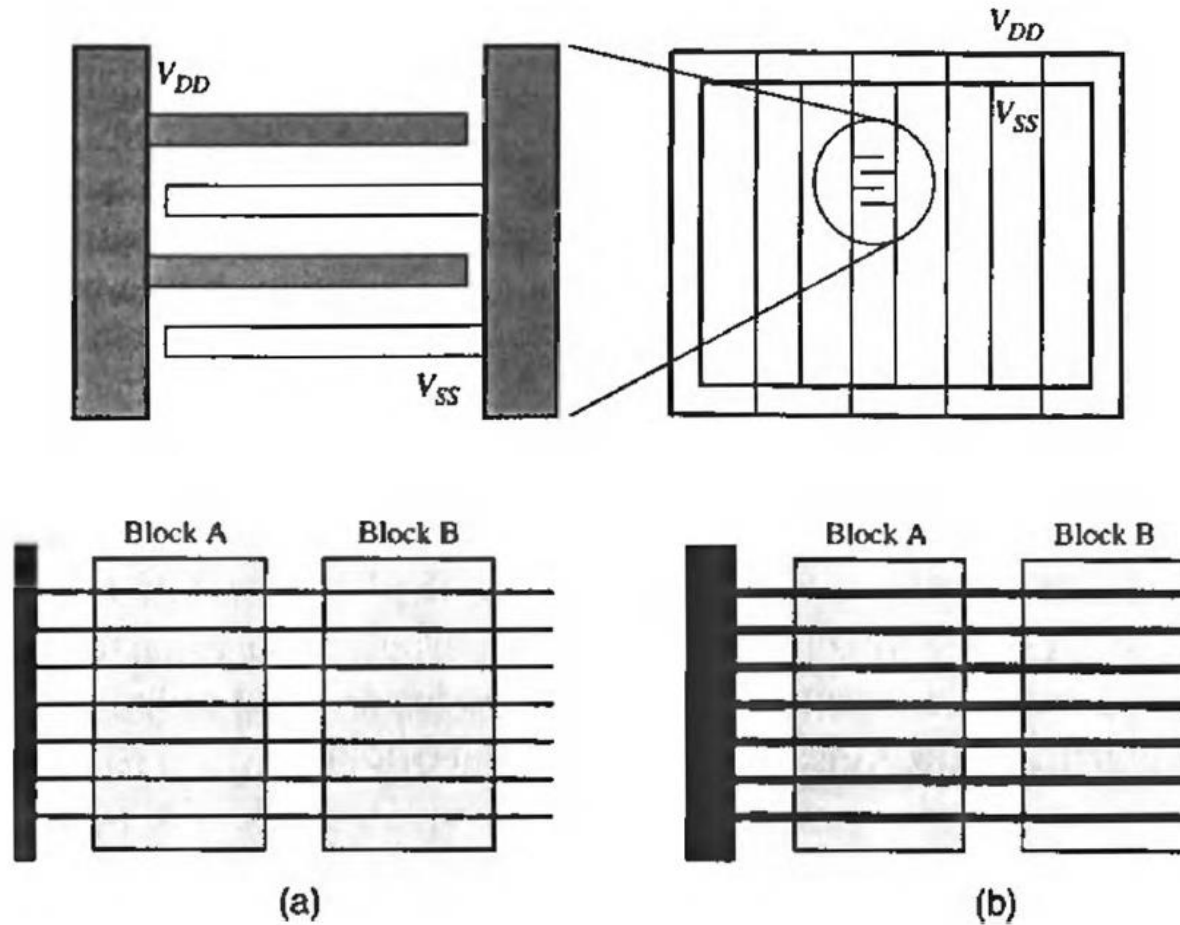
$$t_{50} = \text{MTTF} = 10 \times 10 \text{ years} \times 365 \text{ days} \times 24 \text{ hours} = 8.8 \times 10^5 \text{ hours}$$

Using Equation (11.3), we can solve for $J_{\max}$:

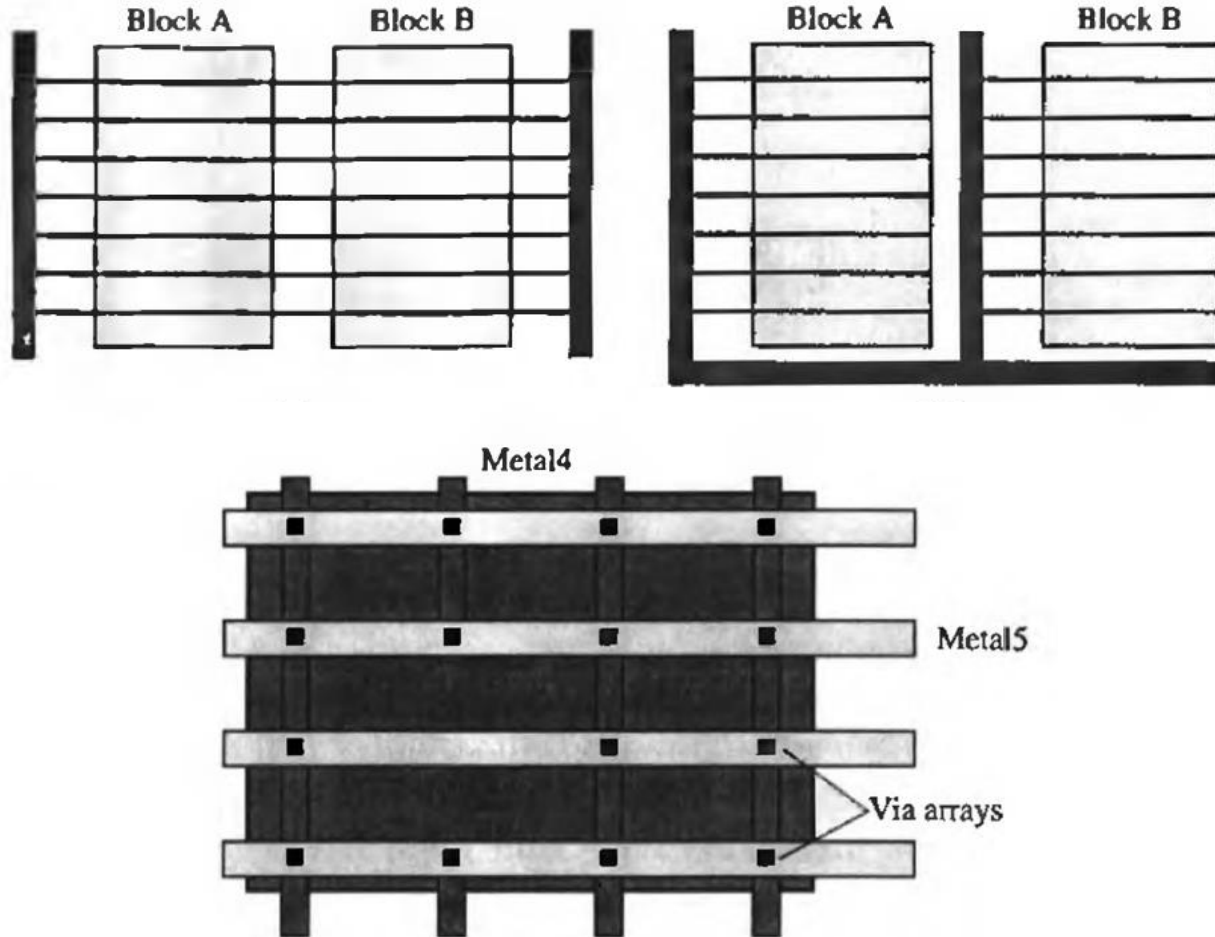
$$\begin{aligned} J_{\max} &= \sqrt{\frac{A}{t_{50}} \exp\left(\frac{\Delta H}{kT}\right)} = \sqrt{\frac{2(10^7)}{8.8(10^5)} \exp\left(\frac{0.85\text{eV}}{8.62 \times 10^{-5}\text{eV/}^\circ\text{K}(398^\circ\text{K})}\right)} \\ &= 8.95(10^5)\text{A/cm}^2 \end{aligned}$$

This value can be used as the maximum tolerable current density for electromigration.

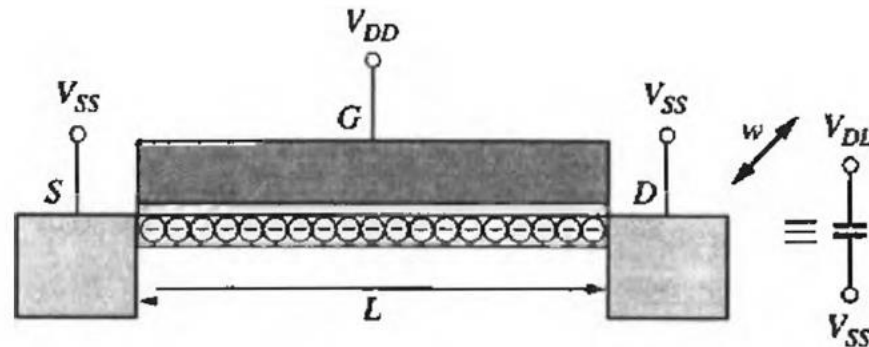
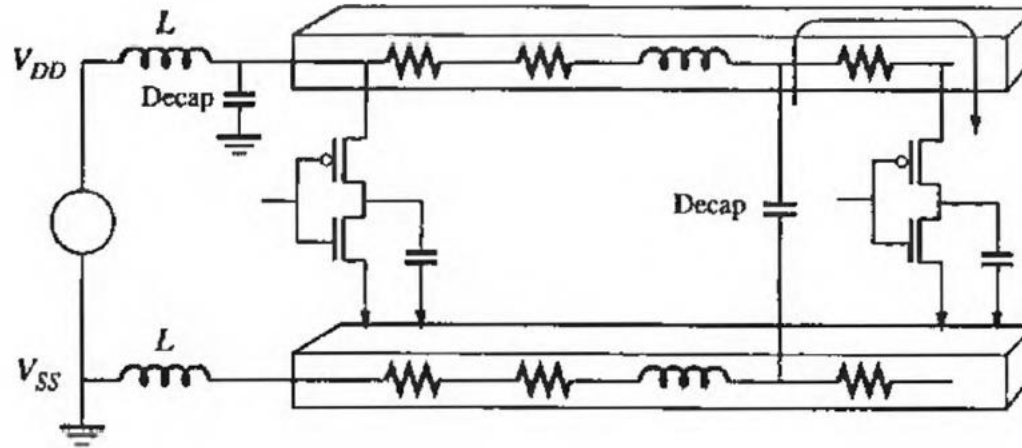
Power Routing Considerations



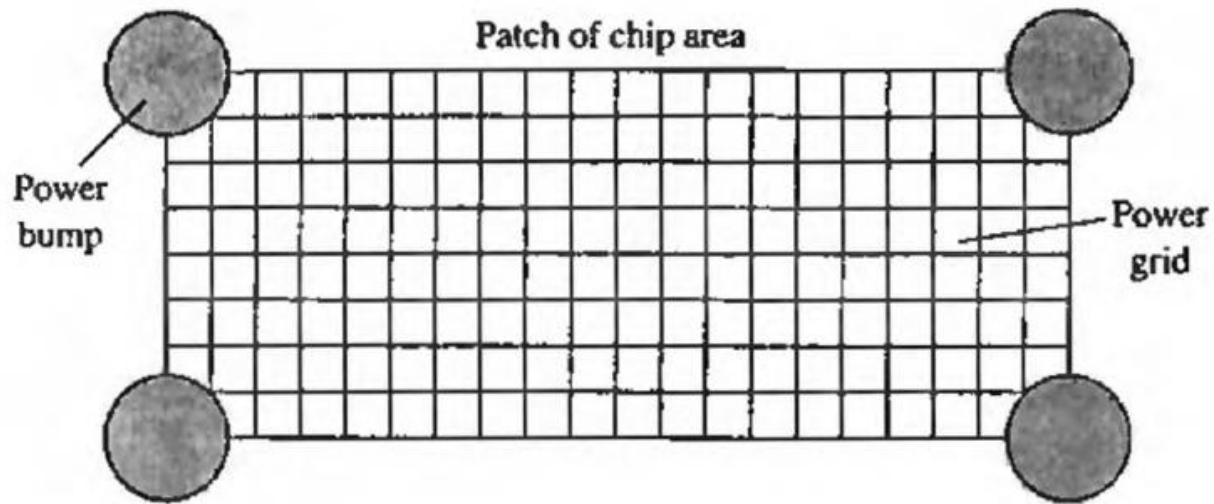
Power Routing Considerations



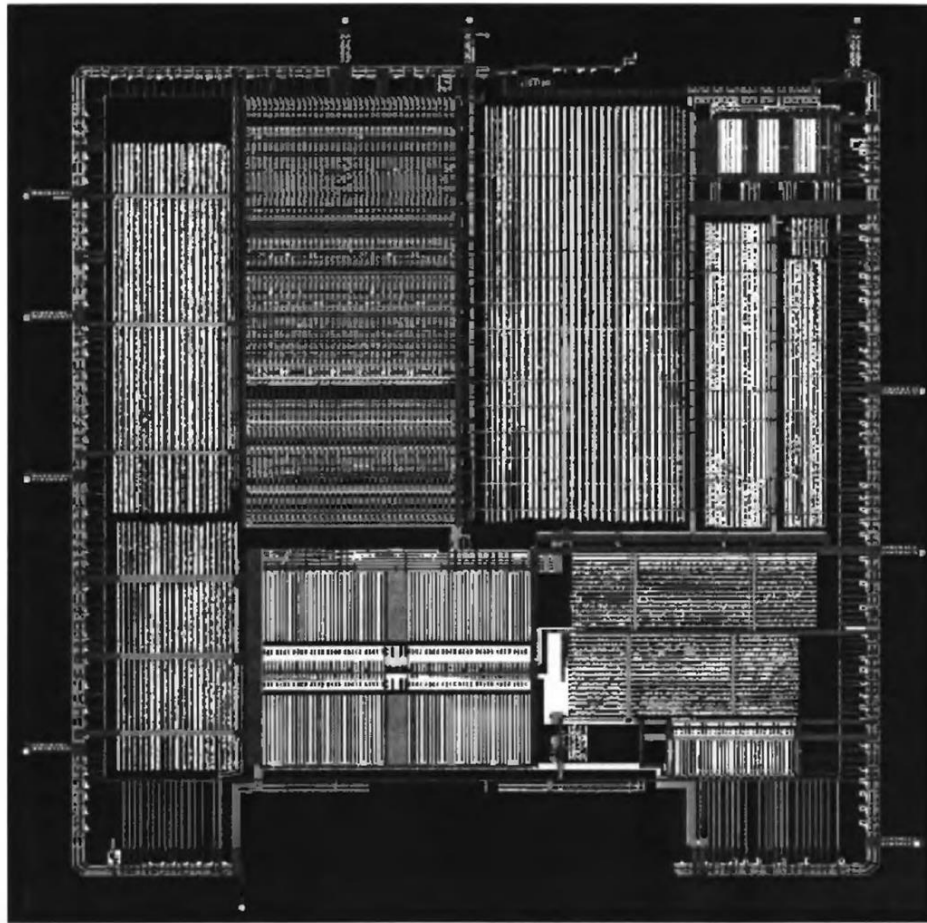
Decoupling Capacitance Design



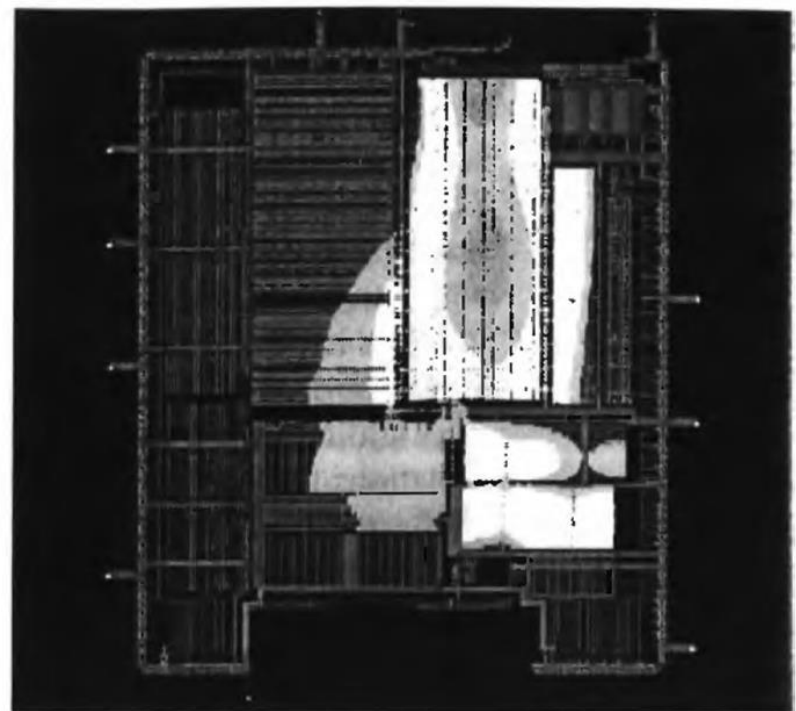
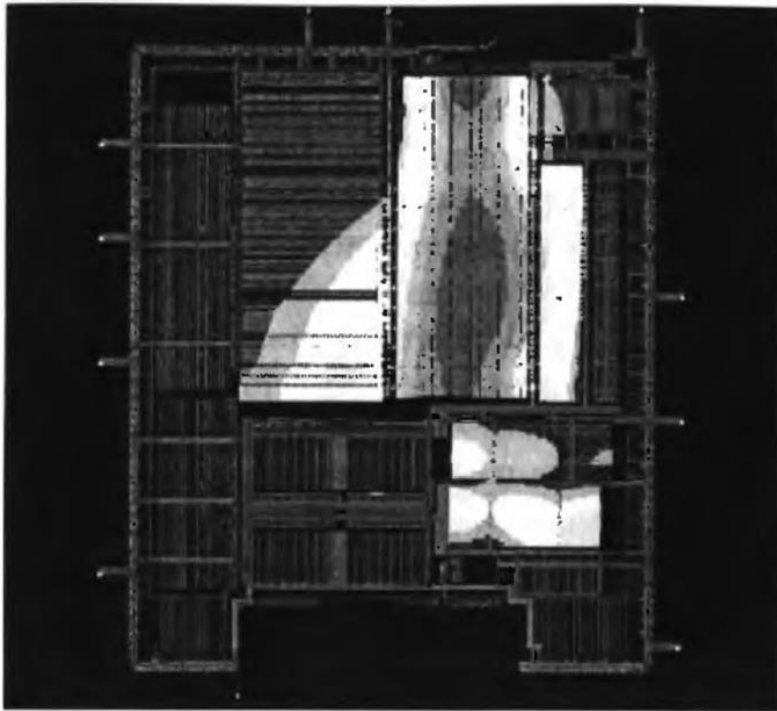
Simulations



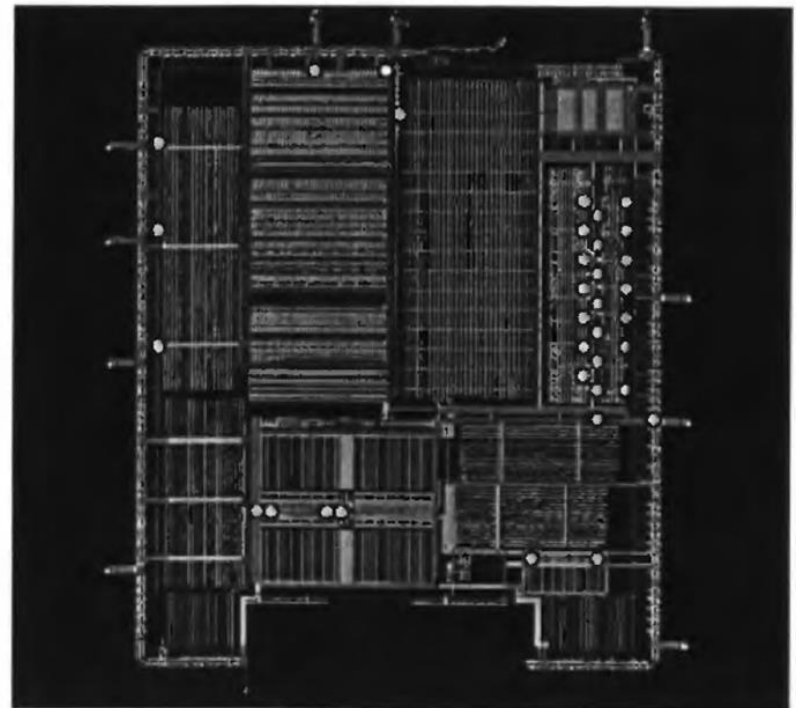
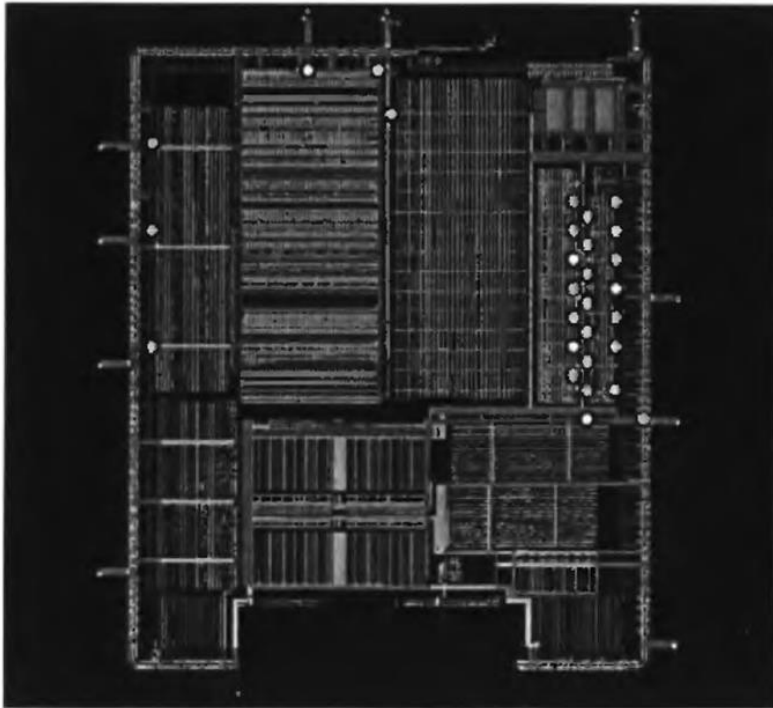
Power Distribution Design Example



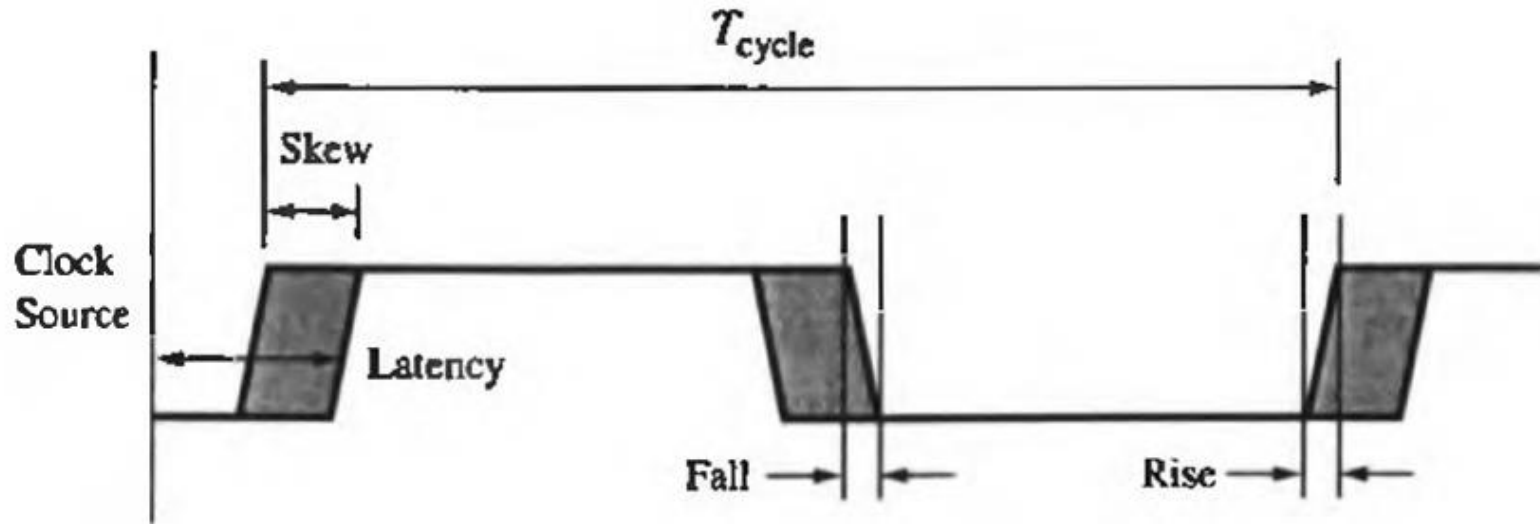
IR Drop Plots



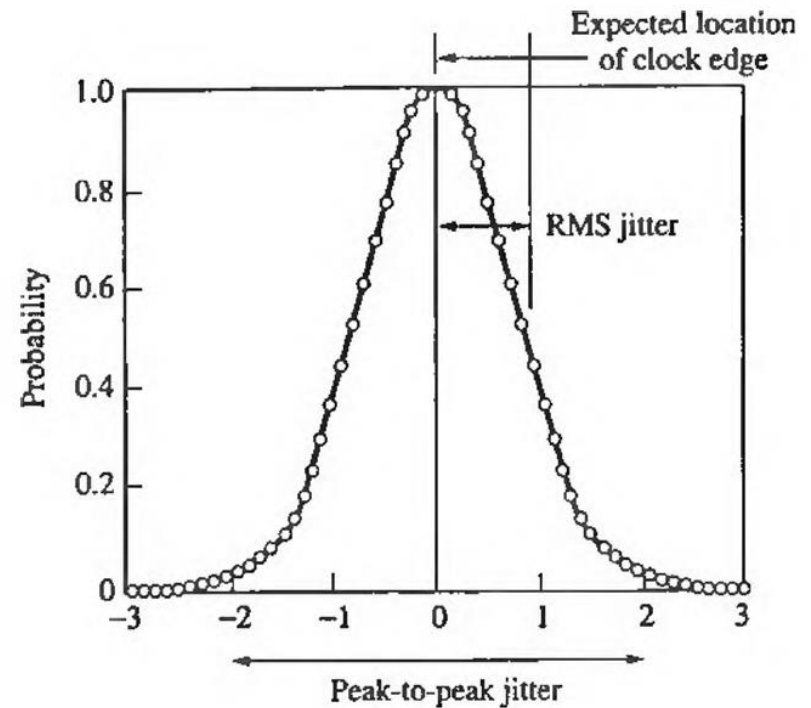
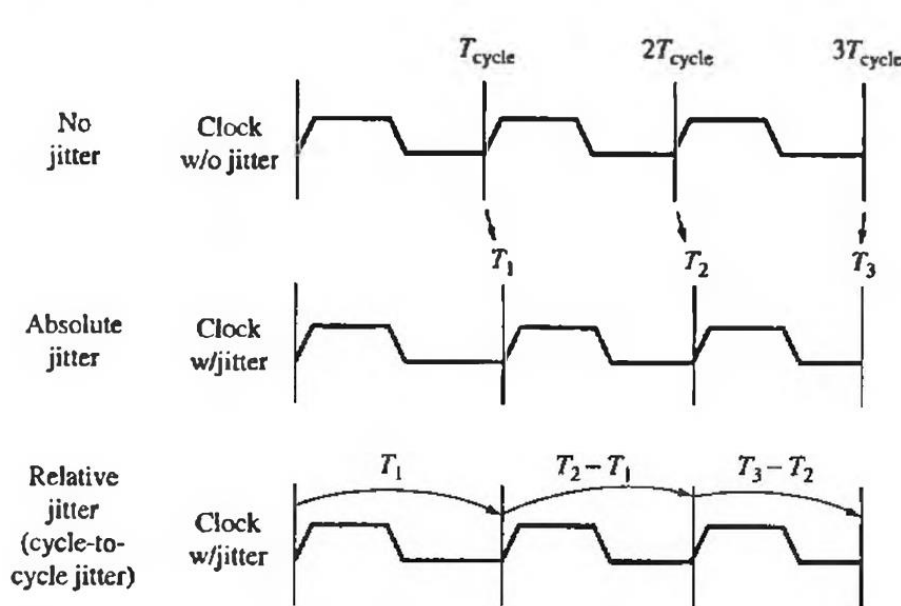
Electromigration Plots



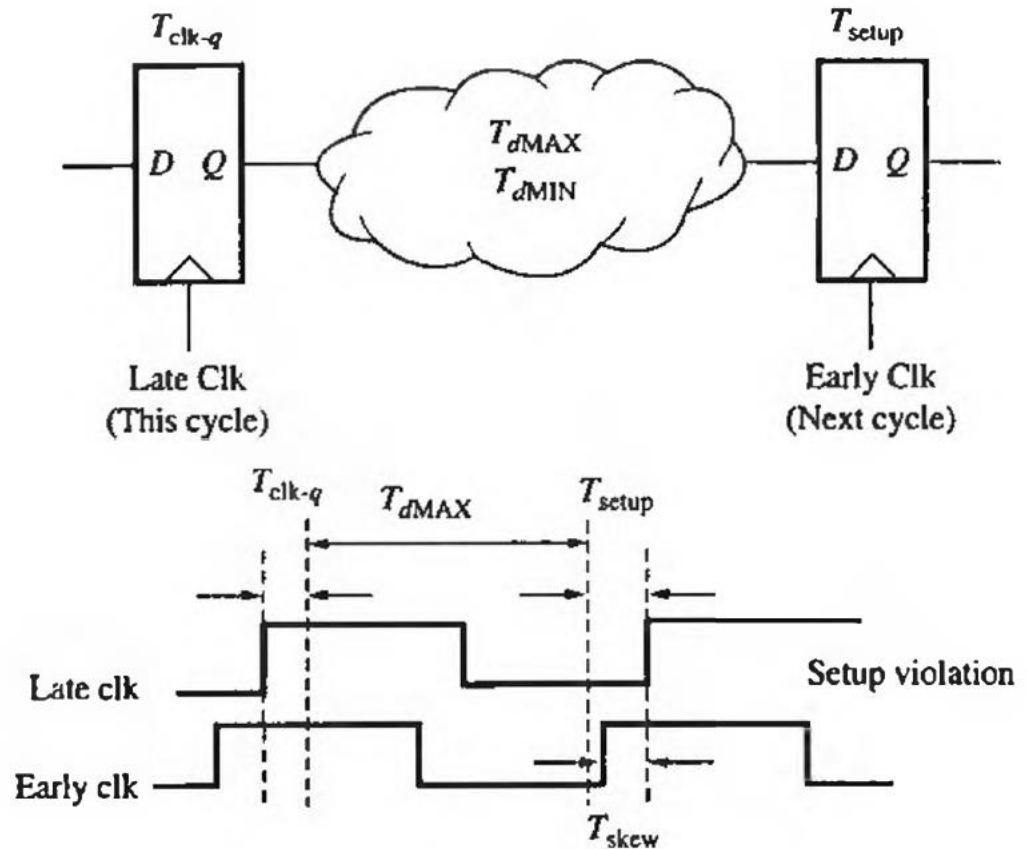
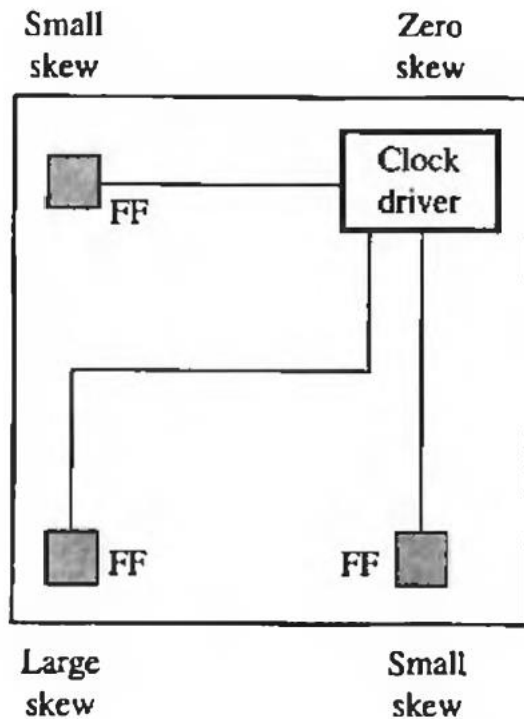
Clocking and Timing Issues



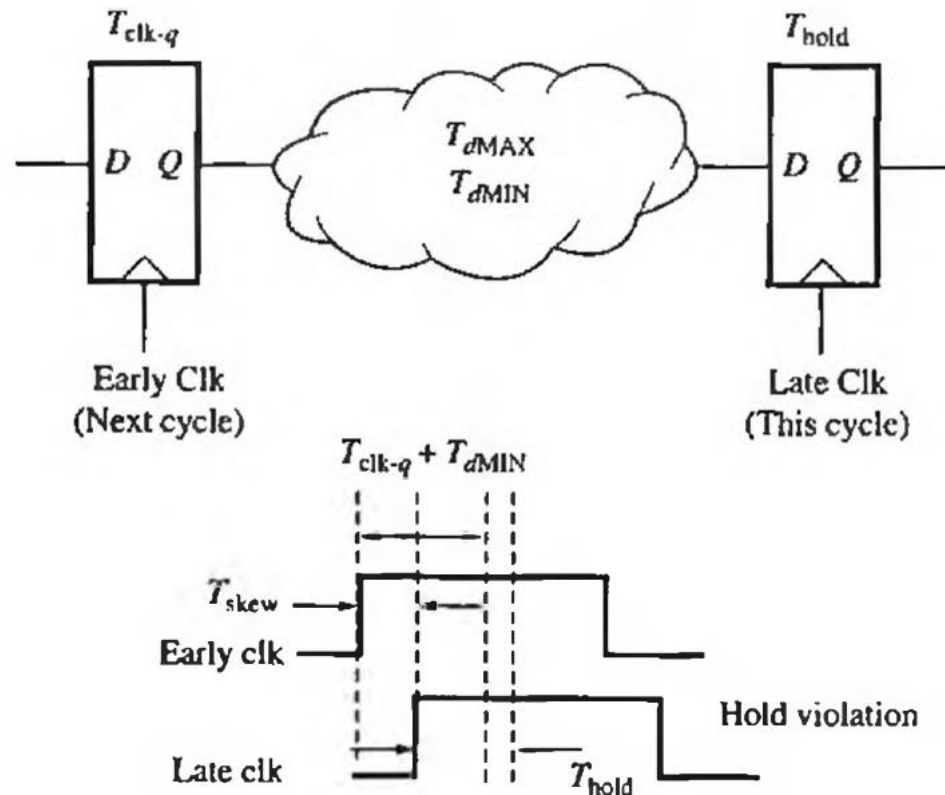
Clock Jitters



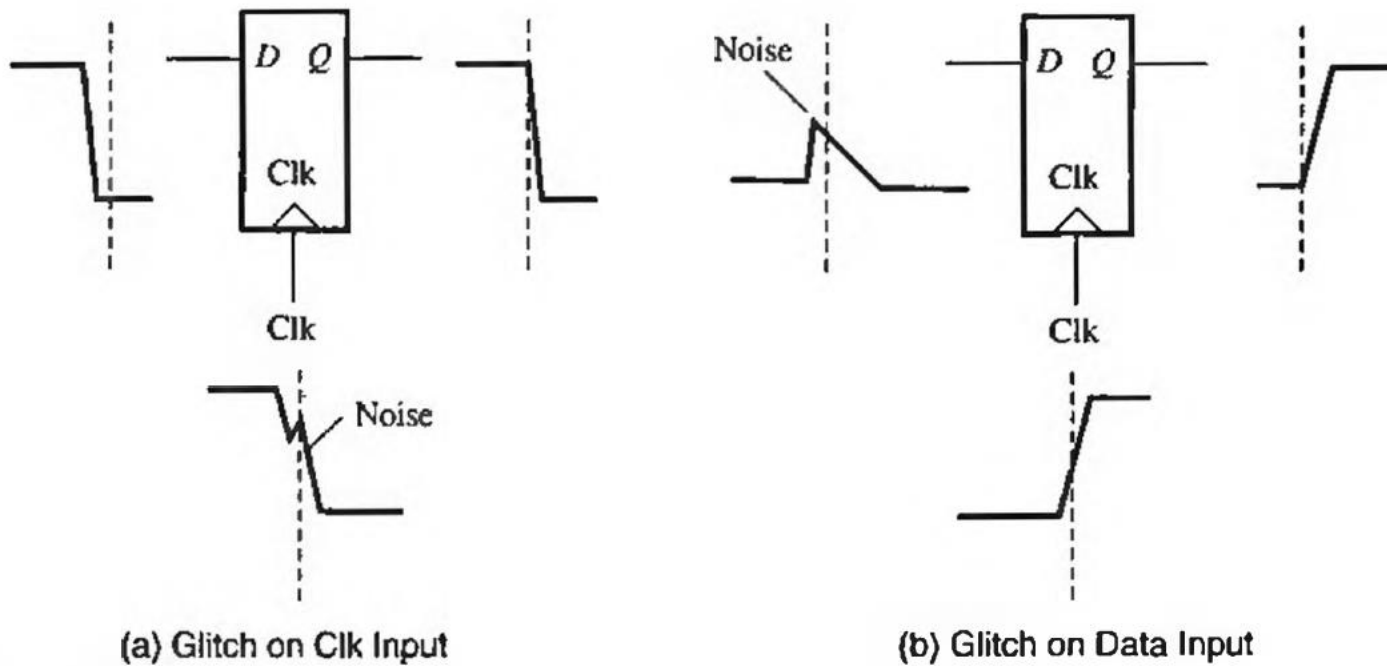
Clock Skew



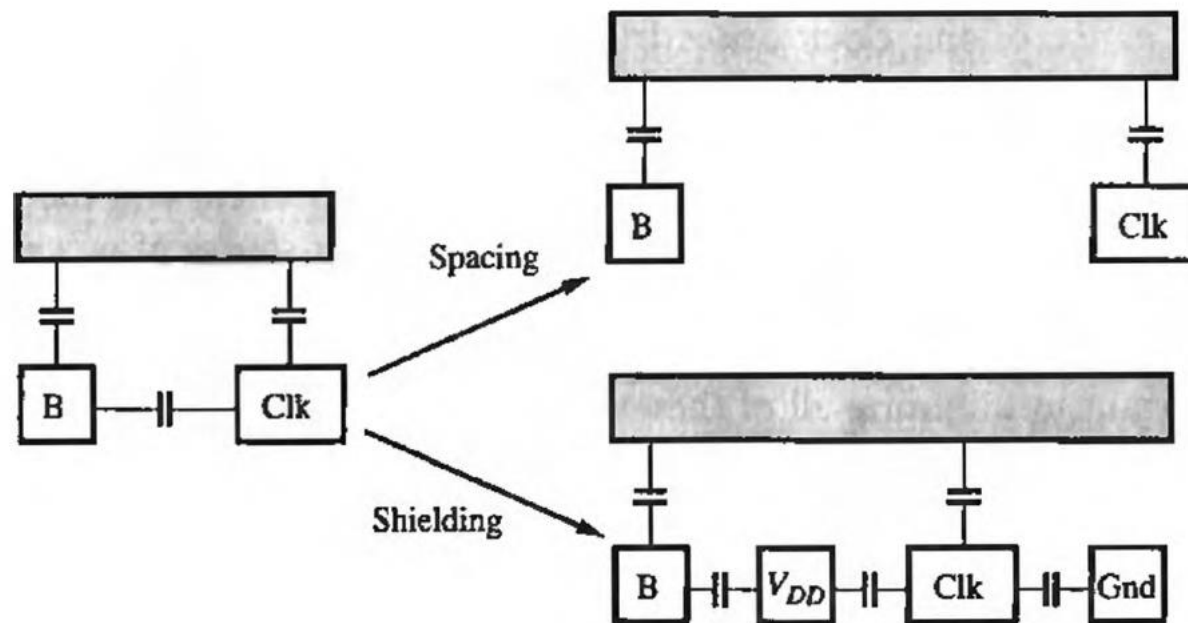
Clock Skew



Noise on Clock and FFs



Layout Solutions to Coupling Problems



Power Dissipation in Clocks

- Switching Power $P = CV_{DD}^2 f_{clk}$

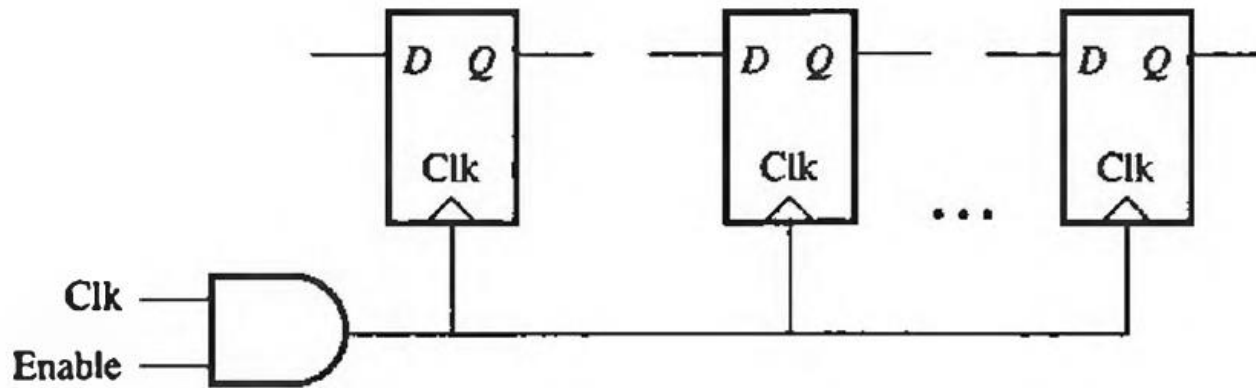
$\alpha = 1$ here, $C \sim \text{nF}$ for large digital circuits

Ex. $P = (1 \text{ nF})(1.8)^2(500)(10^6) = 1.6 \text{ W}$

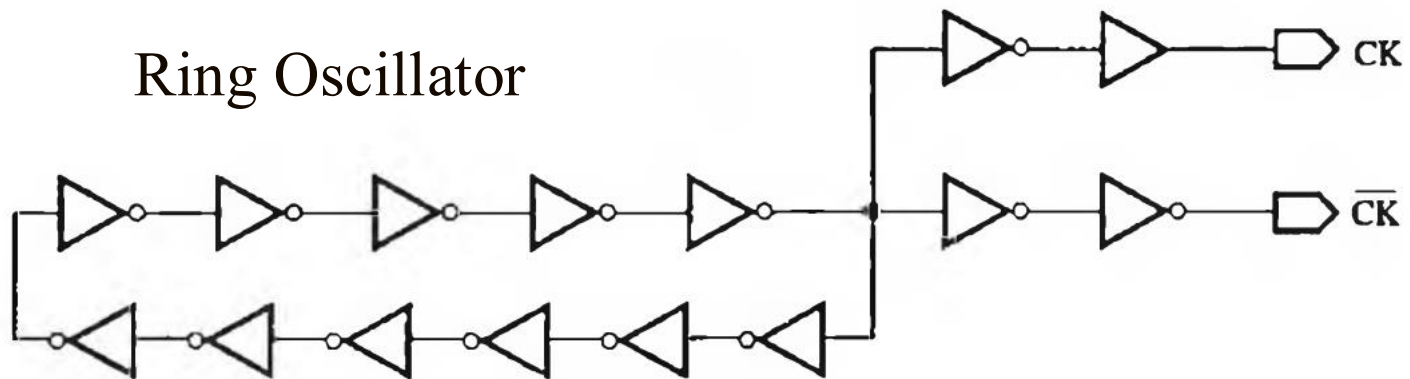
Sources for capacitances:

- Interconnects
- Large buffers
- Inputs to FFs
- Triggering of FFs

Clock Gating

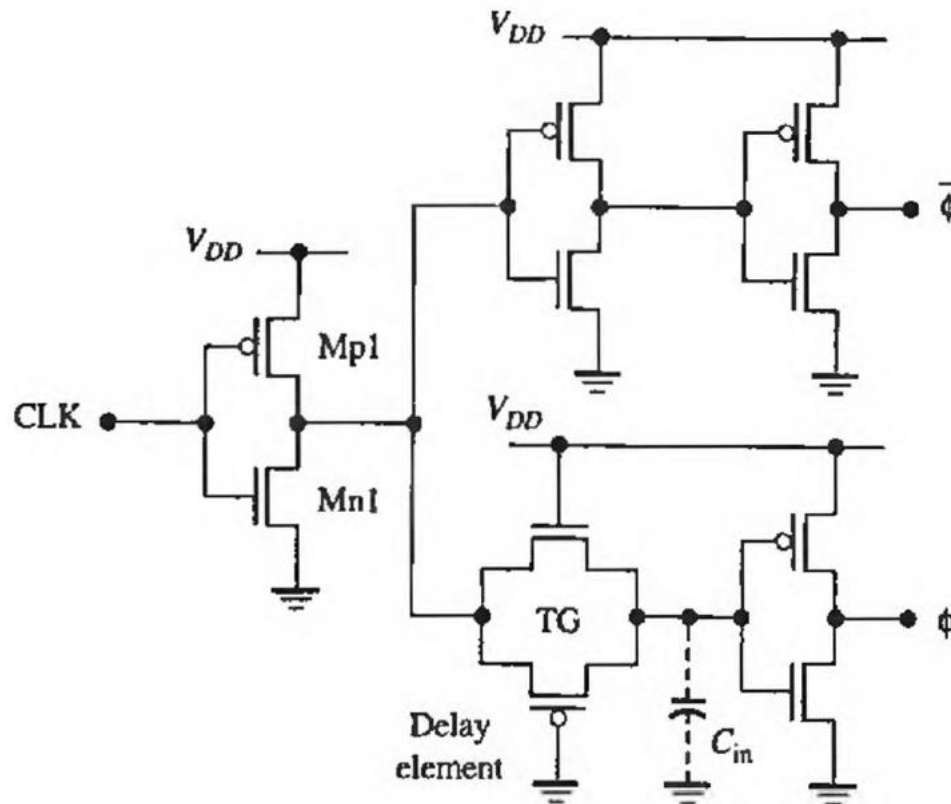


Clock Generation



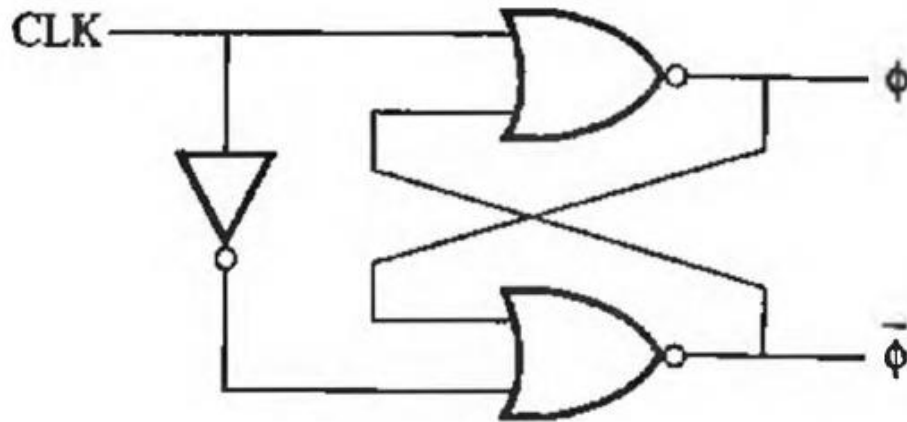
Clock Generation

Clock Generator with TG Delay

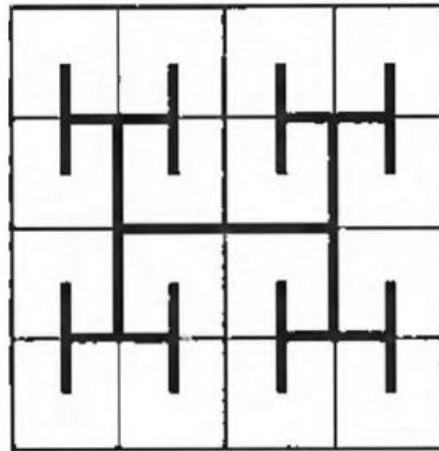


Clock Generation

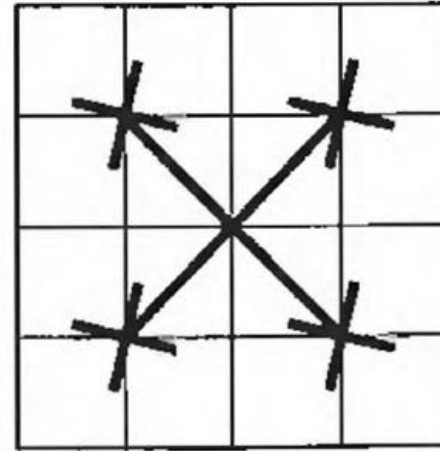
Latch-based Clock Generator



Clock Distribution

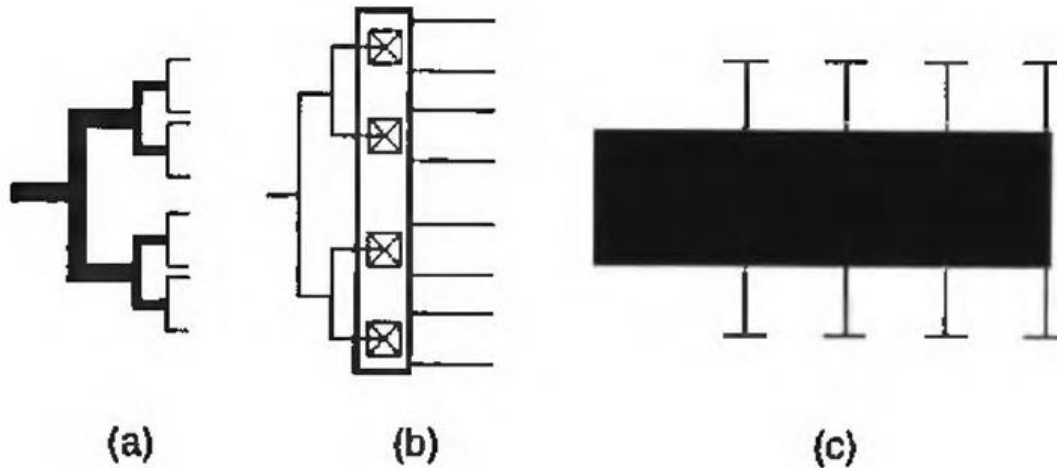


H-tree
(a)

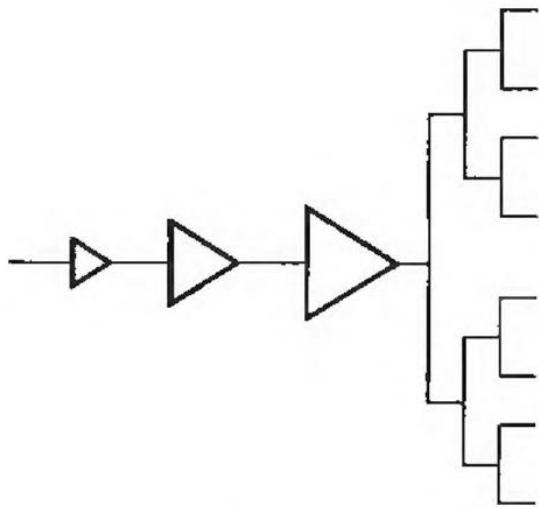


X-tree
(b)

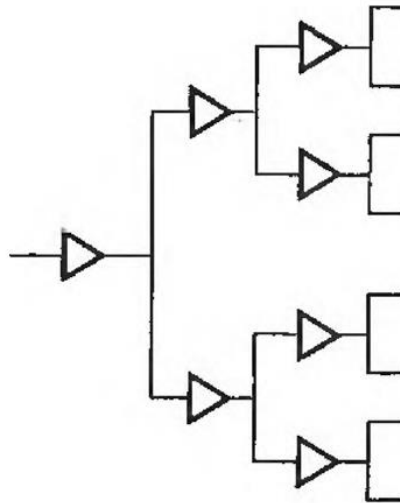
Clock Distribution



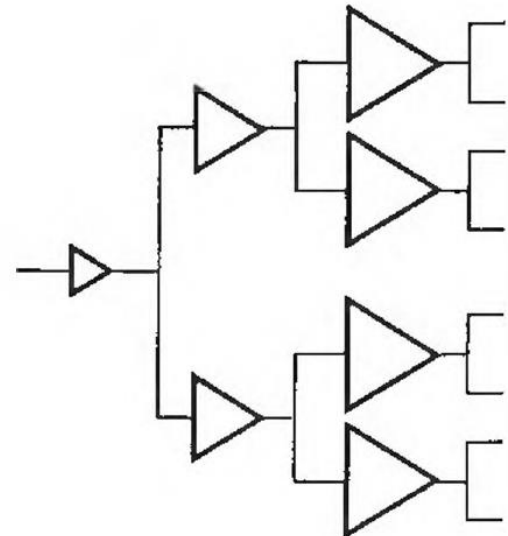
Clock Distribution



(a)

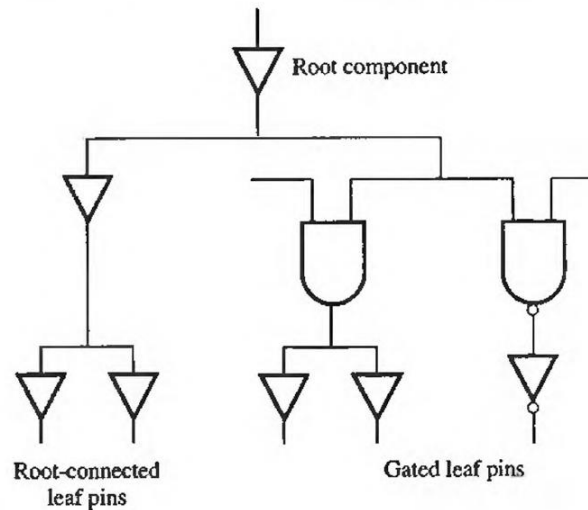
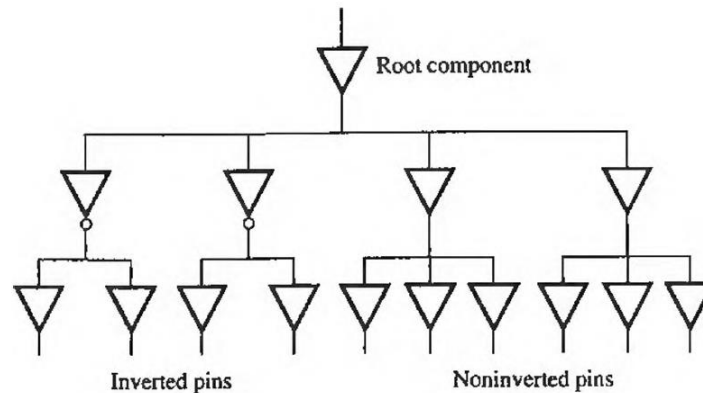


(b)

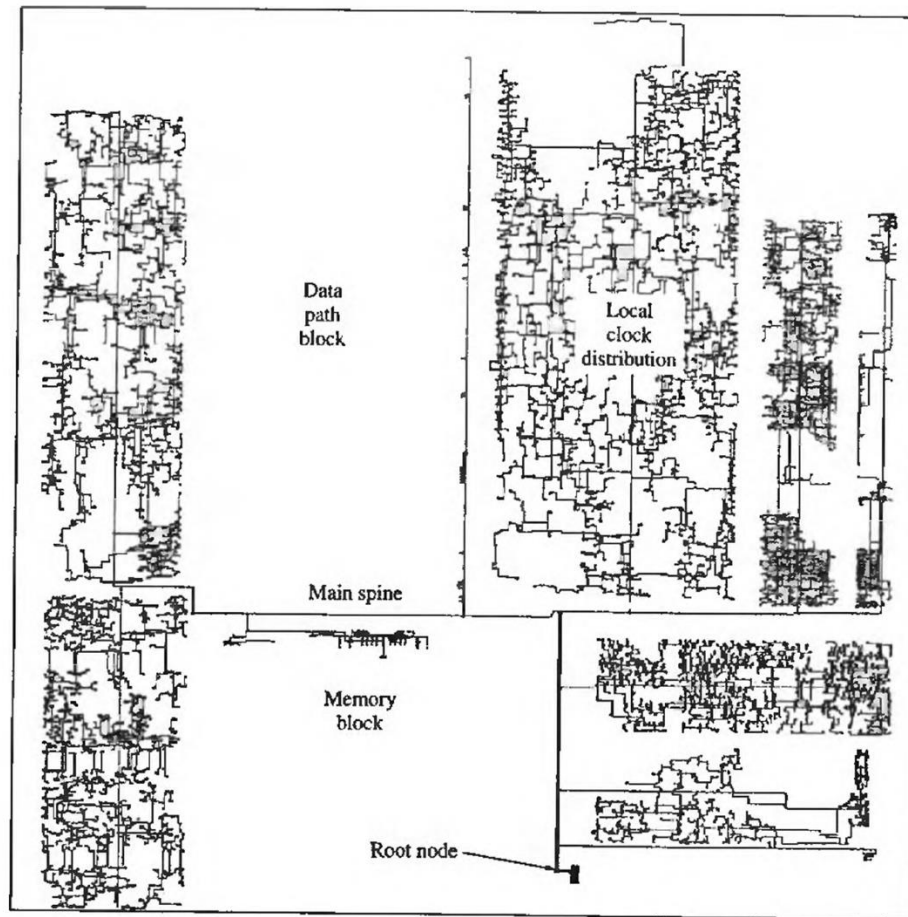


(c)

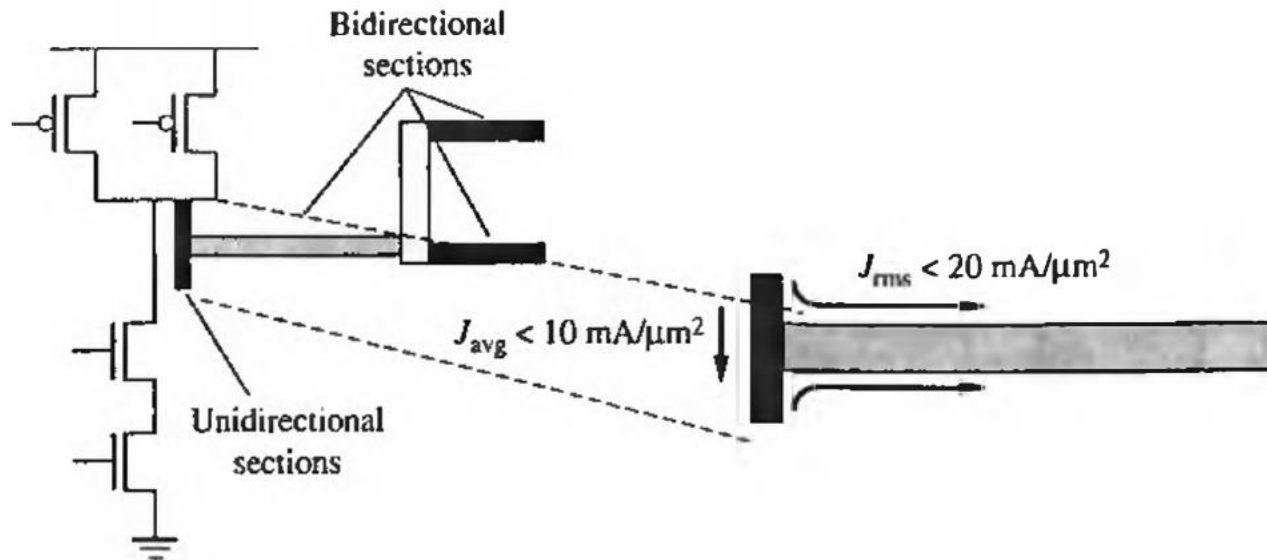
Clock Tree Examples



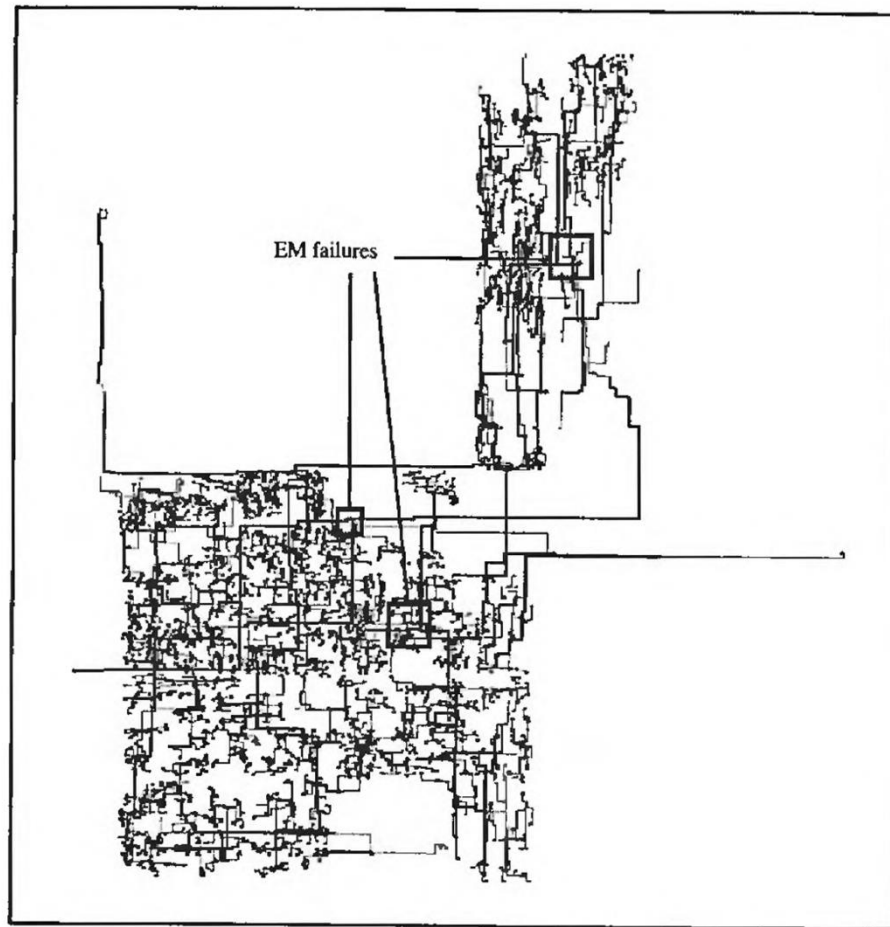
Clock Distribution Network Example



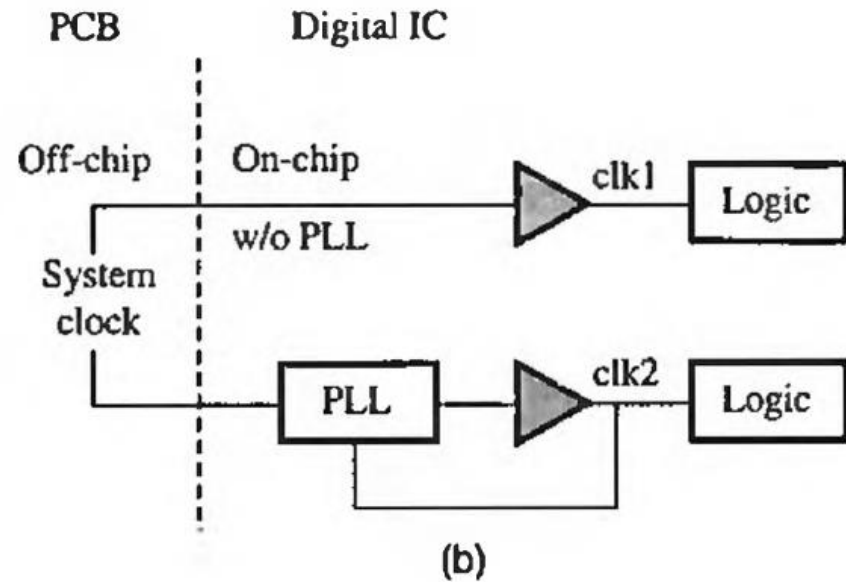
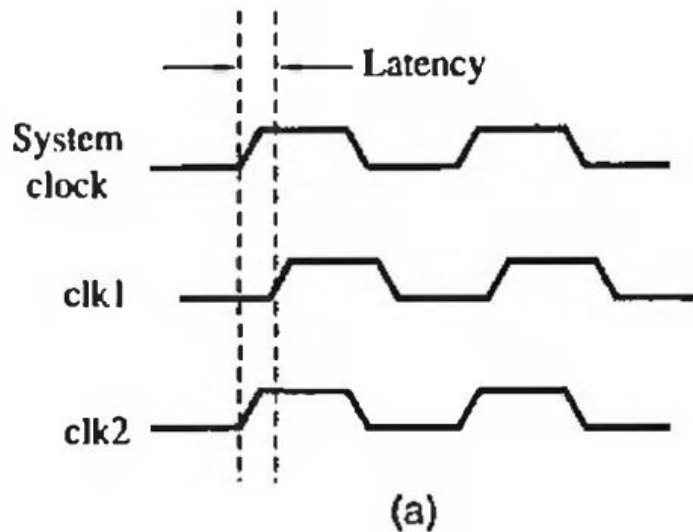
Electromigration Effects



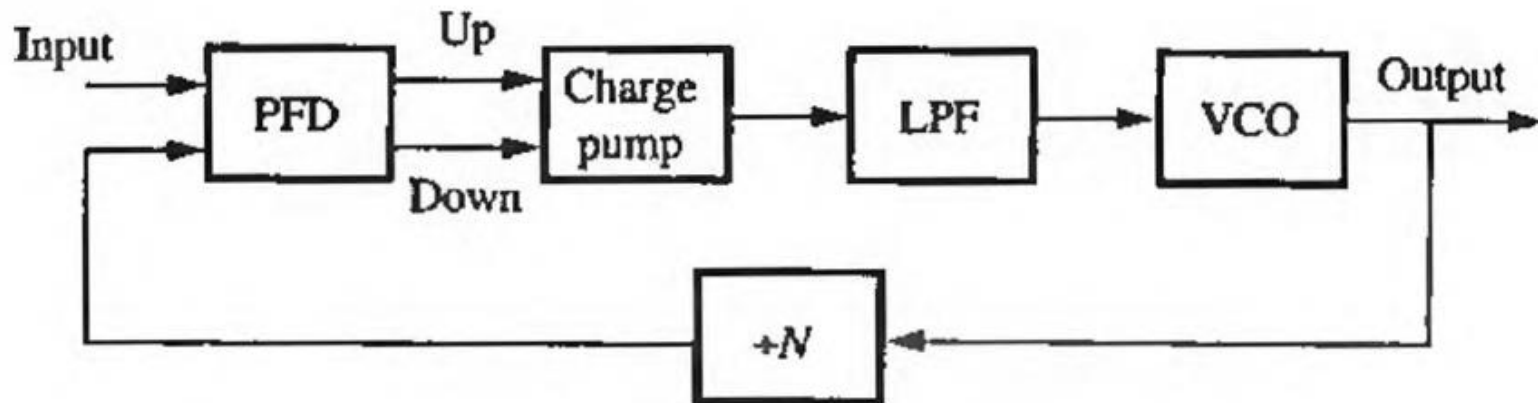
Signal EM Failures in Clock Tree



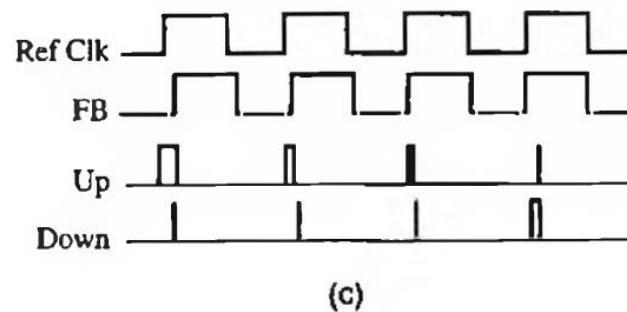
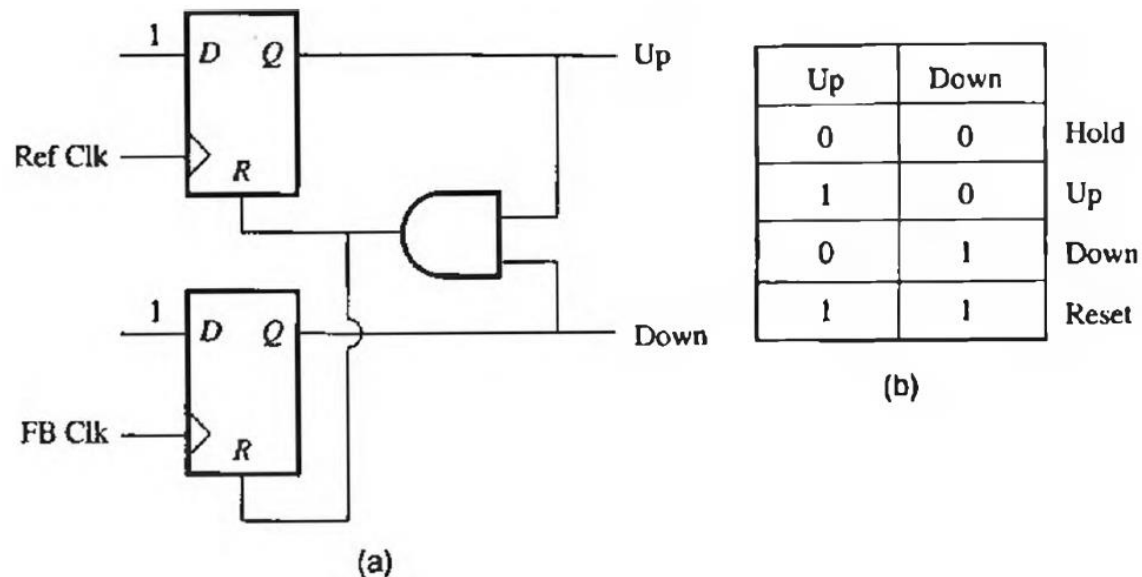
On-Chip Phase Locked Loop



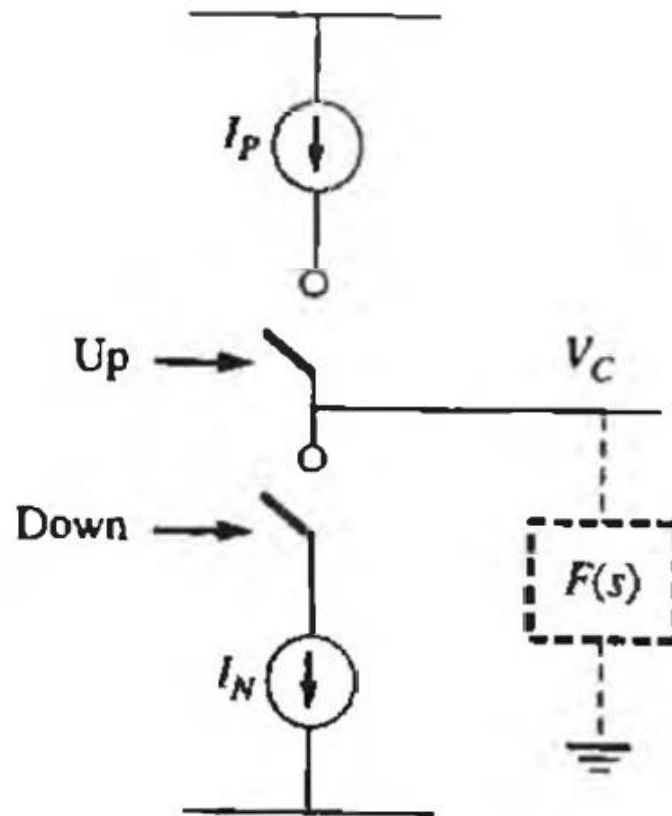
Basic PLL Block Diagram



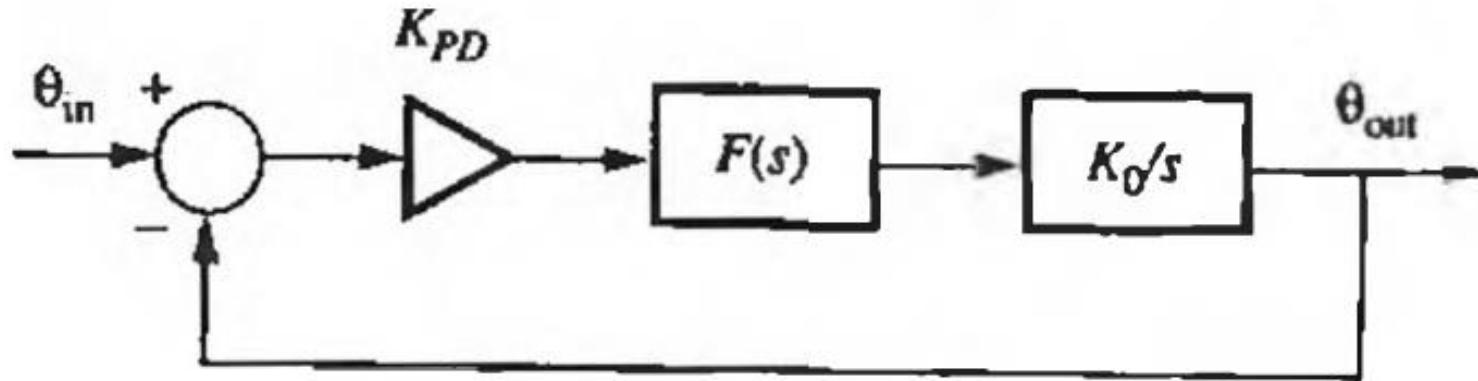
Phase Frequency Detector



Charge Pump and Loop Filter $F(s)$



Linear Feedback Control System for PLL



Open and Closed Loop Gains

We can now determine the two important transfer functions of the system: the open loop gain and the closed loop gain. The open loop gain is obtained by examining the direct path from the input to the output:

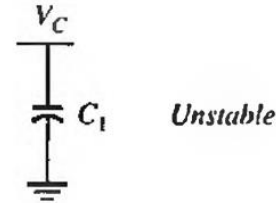
$$\frac{\theta_{\text{out}}}{\theta_{\text{in}}}(s) = K_{PD}F(s) \frac{K_0}{s} = G(s)$$

The closed loop gain is obtained by going around the loop once and can be written in terms of the open loop gain:

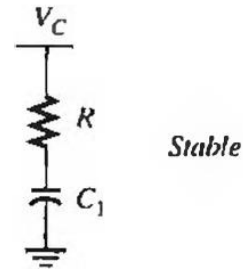
$$\begin{aligned} K_{PD}(\theta_{\text{in}}(s) - \theta_{\text{out}}(s))F(s) \frac{K_0}{s} &= \theta_{\text{out}}(s) \\ \therefore \frac{\theta_{\text{out}}}{\theta_{\text{in}}}(s) &= \frac{K_{PD}F(s) \frac{K_0}{s}}{1 + K_{PD}F(s) \frac{K_0}{s}} = \frac{G(s)}{1 + G(s)} \end{aligned} \tag{11.9}$$

PLL Low-Pass Filter Options

$$G(s) = K_{PD} \frac{1}{C_1 s} \frac{K_0}{s}$$



$$G(s) = K_{PD} \left(\frac{R_1 C_1 s + 1}{C_1 s} \right) \frac{K_0}{s}$$

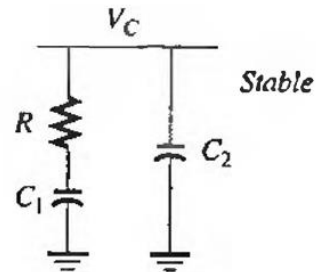


$$G(s) = K_{PD} F(s) \frac{K_0}{s}$$

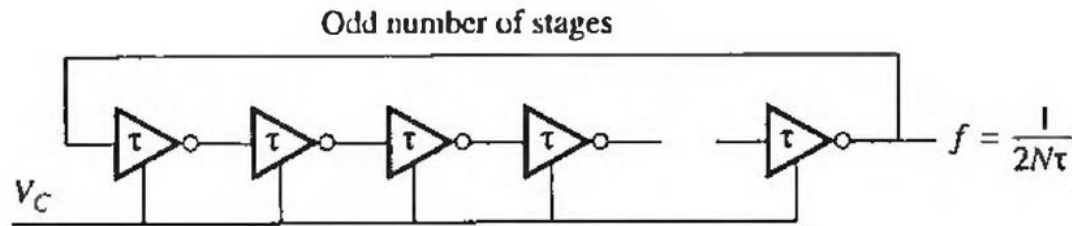
$$F(s) = \left(\frac{b-1}{b} \right) \left(\frac{s\tau_1 + 1}{sC_1 \left(\frac{s\tau_1}{b} + 1 \right)} \right)$$

$$b = 1 + C_1/C_2$$

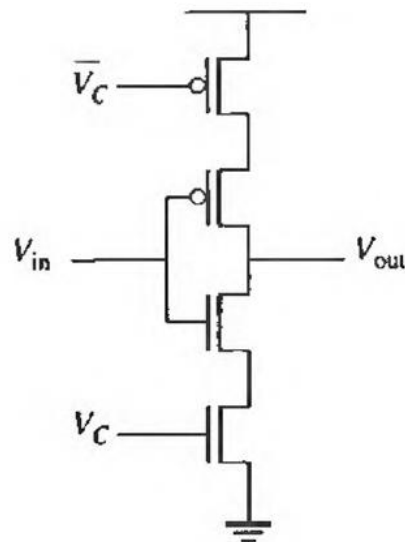
$$\tau_1 = R_1 C_1$$



Voltage-Controlled Oscillator

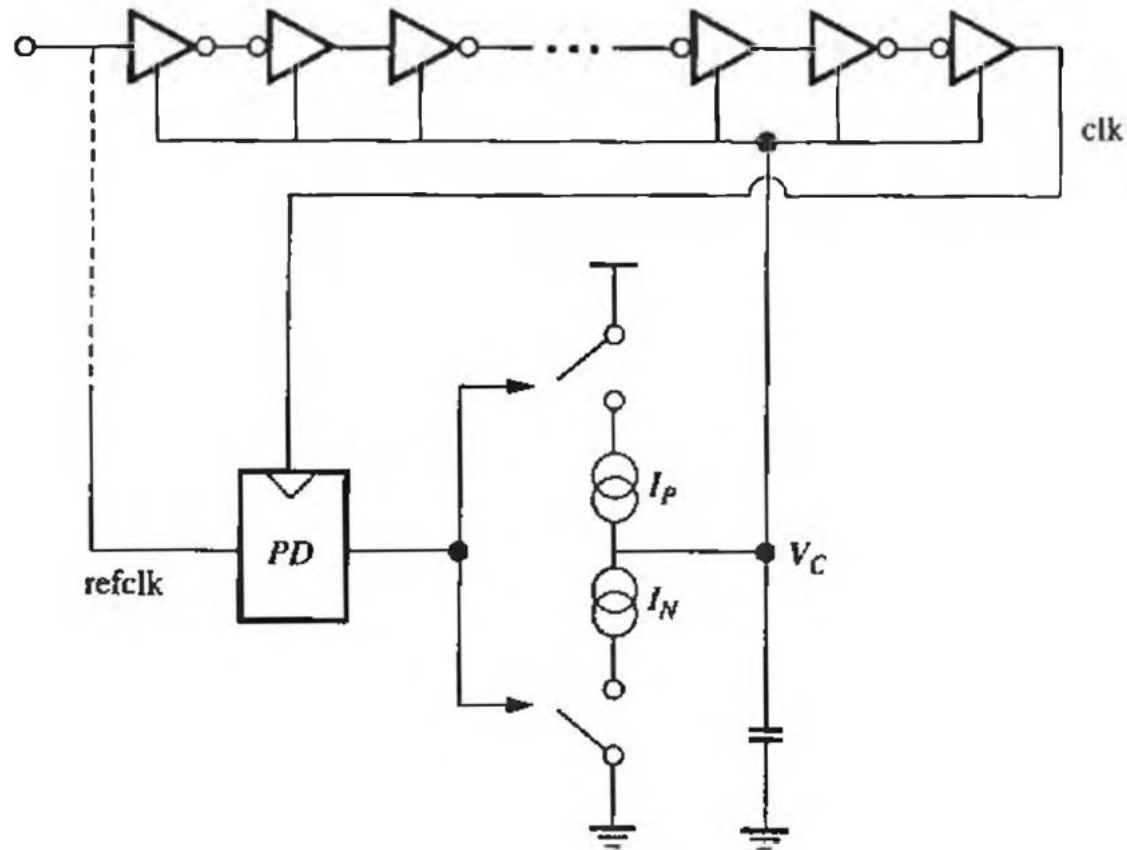


(a)

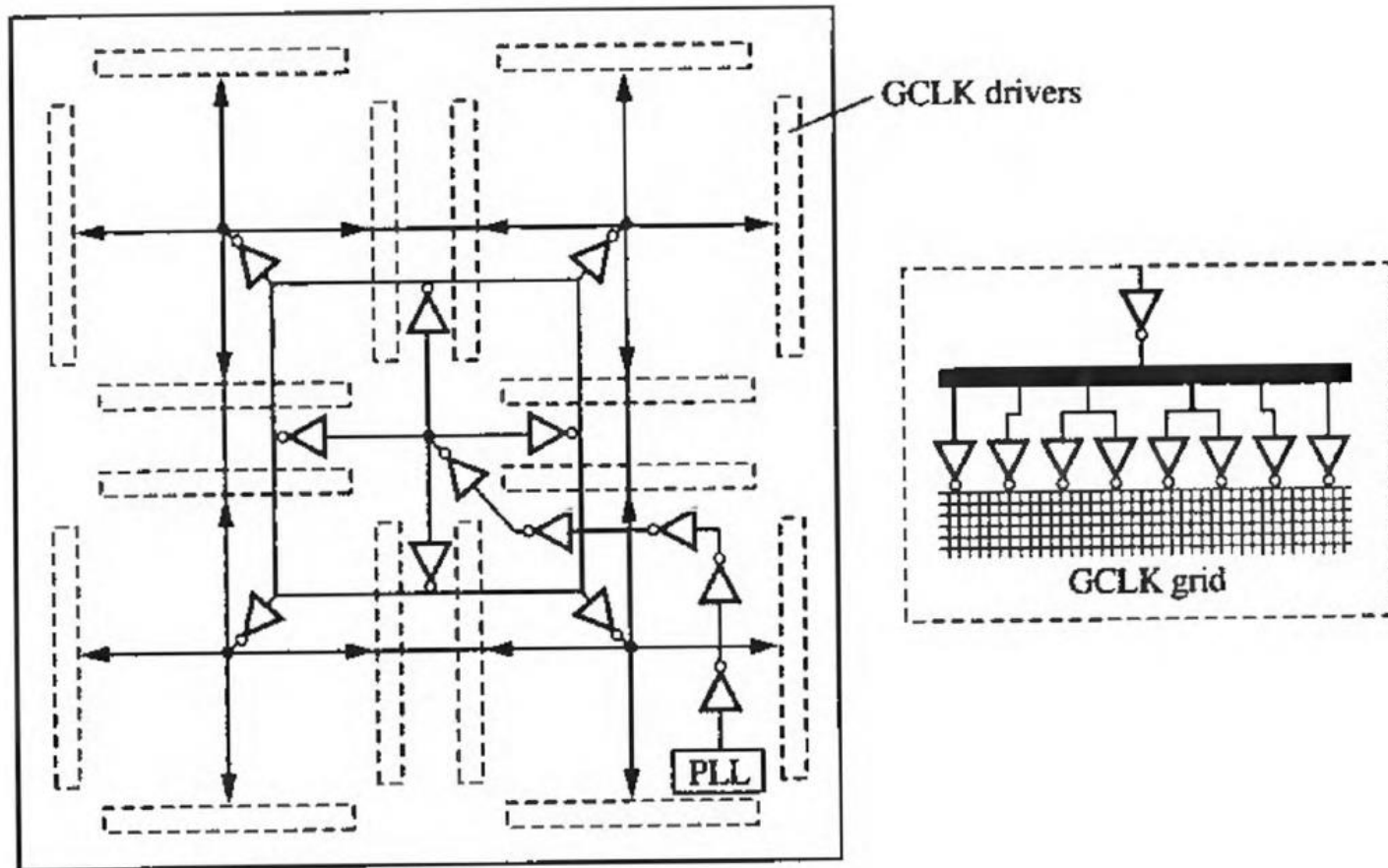


(b)

DLL with VCDL



Global Clock Distribution Network



Clock Hierarchy of a Microprocessor

